

# YANTAO WU

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## EDUCATION

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**Johns Hopkins University, Baltimore, MD**

*Sep 2021 – Present*

Ph.D. Candidate in Mathematics

Advisors: Prof. Mauro Maggioni and Prof. Yannick Sire.

Research Interests: Statistical Machine Learning, LLM Agents, Partial Differential Equations, and AI for Science.

**Syracuse University, Syracuse, NY**

*Jan 2018 – May 2021*

Dual Degree: B.S. in Mathematics and B.A. in Physics

- *Summa Cum Laude* (Cumulative GPA: 3.992/4.000)
- Distinction in Mathematics
- Syracuse University Scholar
- Member of Phi Beta Kappa

## CURRENT RESEARCH PROJECTS

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**Manifold Footprint Model**

*2025–2026*

- Developed a nonparametric regression framework for  $F(x) = f(\Pi_{\mathcal{M}}x)$  that exploits unknown submanifold structure to reduce the effective statistical dimension of high-dimensional regression and mitigate the curse of dimensionality under structural assumptions.
- Designed and implemented CUDA/MEX accelerated cross-validation kernels, including warp-cooperative Jacobi eigensolvers and persistent GPU memory management, achieving roughly  $100\times$  speedup over the MATLAB reference implementation on a 60,000-sample cross-validation task.

**Auto-Research Agent**

*2026*

- Built a multi-agent research automation platform covering literature review, experiment planning, remote execution, result analysis, automated review, and paper/report generation. Deployed across formal verification, LLM compression, PDE foundation model, and particle system generative modeling projects.
- Engineered durable state machines, job-sentinel monitoring, and hook-based workflow enforcement for fault recovery, auditability, and self-healing in long-running experiments.

**AI Verification Agent**

*2026*

- Built LLM-based agents for repository-scale formal verification in Dafny and Lean, aiming to reduce expert manual intervention in safety-critical software verification. Achieved a 51% pass rate on DafnyBench ( $N = 200$ ) under multi-round verifier-guided repair, with ablation studies evaluating retrieval, failure memory, and verifier feedback.
- Integrated proof-relevant retrieval across dependency closures, verifier-guided error feedback, structured failure memory, and failure-type-driven action selection.

**Geometry-Aware LLM Compression**

*2025–2026*

- Analyzed LLM representation redundancy from a high-dimensional submanifold geometry perspective, exploring compression strategies that reduce inference cost while preserving benchmark performance for Qwen and Mistral models from 7B to 32B parameters.
- Built activation-geometry pipelines to analyze hidden-state manifold structure and design layer-wise compression strategies for different Transformer modules, with evaluation on MMLU, GSM8K, HellaSwag, and ARC.

## PDE Foundation Model

2025–2026

- Built Fourier Neural Operator and hybrid neural-PDE architectures for learning PDE dynamics from data, targeting AI-for-Science applications in weather, fluid dynamics, and materials science. Trained and evaluated on PDEBench.
- Investigated the reliability of standard PDE-discovery benchmarks and identified a solver-data mismatch caused by spatial coarsening; used this diagnosis to reduce long-horizon prediction error.

## Interacting Particle System Generative Model

2026

- Developed structured generative world models that recover governing physical laws—single-particle potentials, pairwise interaction kernels, and sparse interaction graphs—from paired endpoint observations of multi-particle systems via structured Schrödinger bridges trained with flow matching.
- Validated the method across synthetic, Coulomb, Lennard-Jones, and molecular dynamics systems; the learned model generalizes to out-of-distribution initial conditions and outperforms standard transport/diffusion baselines in these settings, with applications to molecular dynamics, drug discovery, and collective behavior modeling.

## TECHNICAL SKILLS

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|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ML / AI</b>             | LLM agents, formal verification (Lean, Dafny), model compression, neural operators, generative modeling, flow matching, scientific ML, manifold learning |
| <b>Programming</b>         | Python, PyTorch, CUDA/MEX, MATLAB                                                                                                                        |
| <b>Math &amp; Modeling</b> | Nonparametric regression, statistical learning theory, PDEs, manifold geometry                                                                           |

## PUBLICATIONS AND BOOK CHAPTERS

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- Yantao Wu, Mauro Maggioni. *Conditional Regression for the Nonlinear Single-Variable Model*. *Journal of Machine Learning Research*, revise & resubmit. [arXiv:2411.09686](#)
  - Thomas Y. Hou, Xiang Qin, Yannick Sire, Yantao Wu. *Self-similar blow-up profile for the one-dimensional reduction of generalized SQG with infinite energy*. 2026. [arXiv:2603.11301](#)
  - Fanghua Lin, Yannick Sire, Yantao Wu, Yifu Zhou. *Liquid crystals and topological vorticity: smoothness of mild solutions*. 2026. [arXiv:2601.18726](#)
  - Yannick Sire, Yantao Wu, Yifu Zhou. *Global Existence of Weak Solutions to the Two Dimensional Nematic Liquid Crystal Flow with Partially Free Boundary*. *Journal of the London Mathematical Society*, 2025, 110. [[Paper](#) [arXiv:2308.04358](#)]
  - Yannick Sire, Yantao Wu, Yifu Zhou. *Geometric variational problems with free boundary: harmonic maps, minimal surfaces and applications to a new model of Liquid Crystal Flow*. *Coimbra Mathematical Texts, Modern methods in the analysis of free boundary problems*, [to appear](#).
  - Lee Kennard, Yantao Wu. *Halperin's Conjecture in Formal Dimensions up to 20*. *Communications in Algebra*, 2023, Volume 51, Issue 8. [[Paper](#) [arXiv:2104.04086](#)]

## ACADEMIC AWARDS

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- Chinese Mathematical Olympiad: Provincial First Prize, 2017
  - Chinese Physics Olympiad: Provincial First Prize, 2017
  - William Lowell Putnam Mathematical Competition: ranked in the top 15% nationally in 2018 and 2019
  - Mathematics Department Awards: Euclid Prize, 2020; Archimedes Prize, 2021
  - Physics Department Award: Neil F. Beardsley Prize, 2021